26. Volume & 3D Shapes

Year 4 · Mark Scheme — Supplementary Worksheet

Foundation — Answers

1. 16cm³

2. 6

3. B — cube

4. 12

5. 8

6. 10cm³

7. C — sphere

8. 27cm³

9. A, C and E (a sphere, cylinder and cone all have at least one curved surface; a cube and cuboid are made only of flat faces)

10. 6

11. 12cm³

12. B — a type of cuboid

13. 12

14. 5cm³

15. 8

Standard — Answers

16. 30cm³

17. A — 5×3×2 (5×3×2 = 30cm³; 6×2×2 = 24cm³)

18. 24cm³

19. 5

20. 6

21. B — cylinder

22. 3cm

23. C and E (4×2×2 = 16cm³ and 3×3×2 = 18cm³ are both more than 15cm³; A = 8cm³, B = 12cm³ and D = 10cm³ are not)

24. 8

25. 5

26. 40cm³

27. B — square-based pyramid

28. 64cm³

29. 2

30. 36cm³

31. 4cm

32. 14cm³

33. C — both A and B (4×2×2 = 16cm³ and 8×1×2 = 16cm³)

34. A, B and D (3×2×3 = 18cm³, 2×2×4 = 16cm³ and 4×1×4 = 16cm³ are all more than 15cm³; C = 10cm³ and E = 12cm³ are not)

35. 45cm³

Hard — Answers

36. 3cm (36 ÷ (6 × 2) = 36 ÷ 12 = 3)

37. A, B and D (the box holds 4×3×2 = 24 small cubes, its volume is 24cm³, and each small cube has a volume of 1cm³; C and E are false, since the correct count and volume are both 24, not 18)

38. They are equal (both are 5×2×4 = 40cm³ and 4×5×2 = 40cm³ — multiplying the same three numbers in a different order gives the same volume)

39. 3cm (45 ÷ (5 × 3) = 45 ÷ 15 = 3)

40. B — 3cm (3 × 3 × 3 = 27)

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